5 (a) (i) $Q = CV = 120C\sin(314t)$

(ii)
$$I_C = \frac{dQ}{dt}$$
$$= \frac{d}{dt}\left(120\,C\sin(314\,t)\right)$$
$$= 314\left(120\,C\cos(314\,t)\right)$$
$$= 37700\,C\cos(314\,t)$$

(b) (i)
$$I_{R,rms} = \frac{V_{rms}}{R} = \frac{120/\sqrt{2}}{200} = 0.424\text{ A}$$

(ii)
$$P = \frac{1}{2}P_o \qquad \left(\text{or}: P = I_{rms}{}^2 R\right)$$
$$= \frac{V_o{}^2}{2R}$$
$$= \frac{120^2}{2(200)}$$
$$= 36\text{ W}$$

(c) (i) In a resistor, moving electrons continuously collide with the ionic lattice of the material. Changing the frequency changes how fast the current reverses direction, but it does not change the rate of these collisions. The mean rate of heat generated due to these collisions therefore remains the same, so r.m.s. current through the resistor remains constant.

OR

r.m.s.current through the resistor remains the same, as $I_{R,rms} = \frac{V_o}{\sqrt{2}\,R}$ is constant/ independent of *f*.

(ii) When frequency increases, the capacitor charges and discharges much faster. Because the plates must fill up and empty out in a shorter period, the rate of flow of charge must increase. This increases the r.m.s. current through the capacitor.

OR

r.m.s. current in the capacitor will increase, since $I_{C,rms} \propto \omega \propto f$.